Impact of Health, Environmental, and Animal Welfare Messages Discouraging Red Meat Consumption: An Online Randomized Experiment

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ABSTRACT

Background Reducing red meat consumption is a key strategy for curbing diet-related chronic diseases and mitigating environmental harms from livestock farming. Messaging interventions aiming to reduce red meat consumption have focused on communicating the animal welfare, health, or environmental harms of red meat. Despite the popularity of these 3 approaches, it remains unknown which is most effective, as limited studies have compared them side by side.

Objective Our aim was to evaluate responses to red-meat–reduction messages describing animal welfare, health, or environmental harms.

Design This was an online randomized experiment.

Participants In August 2021, a convenience sample of US adults was recruited via an online panel to complete a survey (n = 2,773 nonvegetarians and vegans were included in primary analyses).

Intervention Participants were randomly assigned to view 1 of the 4 following messages: control (neutral, non–red meat message), animal welfare, health, or environmental red-meat–reduction messages.

Main outcome measures After viewing their assigned message, participants ordered hypothetical meals from 2 restaurants (1 full service and 1 quick service) and rated message reactions, perceptions, and intentions.

Statistical analyses performed Logistic and linear regressions were performed.

Results Compared with the control message, exposure to the health and environmental red-meat–reduction messages reduced red meat selection from the full-service restaurant by 6.0 and 8.8 percentage points, respectively ($P = .02$ and $P < .001$, respectively), while the animal welfare message did not (reduction of 3.3 percentage points, $P = .20$). None of the red-meat–reduction messages affected red meat selection from the quick-service restaurant. All 3 red-meat–reduction messages elicited beneficial effects on key predictors of behavior change, including emotions and thinking about harms.

Conclusions Red-meat–reduction messages, especially those describing health or environmental harms, hold promise for reducing red meat selection in some types of restaurants. Additional interventions may be needed to discourage red meat selection across a wider variety of restaurants, for example, by making salient which menu items contain red meat.
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Average consumption of red meat (ie, beef, pork, lamb, or veal) in the United States remains well above recommended levels.[1,2] High consumption of red meat contributes to cardiovascular disease,[3,4] type 2 diabetes,[3,5] and some cancers.[3,6-10] Red meat production is also a major source of environmental harms, including greenhouse gas emissions,[3,11,12] biodiversity loss,[12] water pollution,[3,11] and deforestation.[12,13] Reducing red meat consumption is therefore an important avenue for improving population health and mitigating environmental damage from food production.

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Messaging interventions offer a promising, scalable approach for reducing red meat consumption. Messaging interventions, including front-of-package labels, product promotions, and media campaigns can lead to meaningful changes in purchases of targeted products, including sugary drinks, low-fat milk, fruits and vegetables, and alcohol.[14-19] Messaging interventions can also be tailored to specific populations and widely disseminated across a variety of settings[17,20-29] and may be more feasible to implement than taxes or bans.[30-33]

Messaging interventions aiming to reduce red meat consumption have adopted varying approaches. Some focus on appealing to concerns about animal welfare,[34] for example, by depicting the suffering of animals raised as livestock.[35] Others focus on the effects of meat consumption on health,[36] for example, by communicating about red meat consumption and heart disease[37] or diabetes.[38] A third approach is to focus on environmental sustainability,[36] for example, by linking meat production to environmental destruction.[39,40] Despite the popularity of these approaches, it is unclear which is most effective, as the small number of studies that have compared 2 or more of these strategies have yielded mixed results.[41-47]

To inform efforts to reduce red meat consumption, we evaluated how red-meat–reduction messages focused on animal welfare, health, or the environment affected restaurant meal choices. Restaurants are an important source of red meat[1,48] and restaurant meals are often served in larger portions than meals prepared at home[49]; shifting consumers away from red meat in restaurants therefore has the potential to meaningfully reduce population-level red meat consumption.

RESEARCH SNAPSHOT

Research Question: Do messages linking red meat consumption with animal welfare, health, or environmental harms discourage selection of red meat?

Key Findings: In a randomized experiment, exposure to red-meat–reduction messages about health or environmental harms reduced hypothetical selection of menu items containing red meat from a full-service restaurant by 6.0 and 8.8 percentage points, respectively, compared with exposure to a neutral control message ($P = .02$ and $P < .001$, respectively). The animal welfare message did not significantly reduce the likelihood of selecting a menu item containing red meat from the full-service restaurant (reduction of 3.3 percentage points compared with the control message; $P = .20$). None of the red-meat–reduction messages affected hypothetical selection of red meat from a quick-service restaurant.

METHODS

Participants

A national sample of US adults (18 years or older) was recruited through CloudResearch Prime Panels, a survey research company that aggregates multiple online panels to provide access to millions of study participants. Compared with participants on the crowdsourcing platform Amazon Mechanical Turk, participants in the Prime Panels pool tend to be more diverse in age, family composition, education, and political attitudes, as well as more representative of the United States.[50] Participants were recruited via e-mail invitations that included generic information on survey length and incentives (ie, no information about the study's purpose, eg, "There is a new study you qualify for!"; the exact text varied by panel). Recruitment occurred from August 13 to August 24, 2021, when the target sample size was achieved (see Analysis section). The design and analytic plan were registered before data collection (https://aspredicted.org/KSR_Q2R; deviations are described in the Analysis section).

Procedures

Participants provided informed consent and completed an online survey programmed in Qualtrics[XM] (median completion time was 13.1 minutes).[51] Qualtrics[XM] assigned participants using simple randomization to 1 of the 4 following message groups: control, animal welfare, health, or environmental. In the control group, participants viewed a control message encouraging reading (a neutral topic unrelated to red meat). In the animal welfare, health, and environmental groups, participants viewed red-meat–reduction messages linking red meat consumption with their assigned topic. All messages used similar syntax and length. Use of a neutral control message controlled for exposure to any message, similar to prior studies.[52–56] A professional graphic designer prepared the messages to mimic social media messages (Figure 1).

Participants viewed their randomly assigned message and completed 2 restaurant ordering tasks: 1 from a popular full-

Figure 1. Control, animal welfare, health, and environmental messages used in the online randomized experiment.

service restaurant (ie, with table service; Chili's) and 1 from a popular causal, quick-service restaurant (ie, without table service; Panera), presented in random order. This study examined both full- and quick-service restaurants because consumer choices may differ in these settings.[54,57–59] Participants viewed their assigned message before each ordering task (ie, 2 exposures). Participants in the red-meat–reduction message groups also viewed a definition of red meat before their first message exposure.

For both ordering tasks, participants viewed a modified menu from the restaurant's online ordering webpage (Figure 2; available at www.jandonline.org). Menus displayed entrées only, given that other types of items (eg, desserts) rarely contain red meat and would be unlikely to be affected by the intervention. Neither menu displayed the restaurant name. The full-service restaurant offered 20 items, including 8 containing red meat (of which 6 were red-meat–focused items [ie, red meat was the primary ingredient, eg, beef burger]), 6 containing poultry or fish but no red meat (eg, chicken fajitas), and 6 vegetarian items (eg, cheese pizza). The quick-service restaurant offered 19 items: 5 containing red meat (of which 1 was a red-meat–focused item [steak sandwich]), 7 containing poultry or fish but no red meat (eg, turkey chili), and 7 vegetarian items (eg, grilled cheese). Menus displayed prices and calorie content for all items as collected from restaurants' websites in 2021.

For both ordering tasks, participants were instructed to imagine that they were ordering a meal for themselves and to select 1 item they would like to eat. To incentivize participants to select items they wished to eat, the survey indicated that 25 participants would be chosen at random to have 1 of their selections from the 2 ordering tasks delivered to them or to receive an electronic gift card of equivalent value (instructions did not specify a retailer for the gift card).

After completing the ordering tasks, participants responded to survey questions. CloudResearch sent participants previously agreed upon incentives (eg, reward points redeemable for gift cards) determined based on survey length and participant characteristics. Study procedures were approved by the Harvard Longwood Campus Institutional Review Board (#21-0568).

Measures

The primary outcome was selection of an item containing red meat in the restaurant ordering tasks. Secondary outcomes included calories, saturated fat, sodium, sugar, and cholesterol selected in the ordering tasks.

The survey also assessed participants' reactions to their assigned messages, including perceived message effectiveness[60]; negative emotional reactions[61]; thinking about the animal welfare, health, and environmental harms of red meat[62]; and message reactance.[63] In addition, the survey assessed perceptions that limiting red meat intake reduces harms,[64,65] perceptions that eating red meat contributes to harms,[66] and intentions to limit red meat consumption.[67] Table 1 (available at www.jandonline.org) shows survey items and response options.

The survey ended with a debrief about the study's purpose and informed participants that those randomly chosen for the bonus incentive would receive a gift card of value equivalent to or greater than their meal selection. Participants could ask that their data be removed.

Analysis

Analyses examined whether animal welfare, health, or environmental red-meat–reduction messages reduced selection of red meat compared with a control message. No specific predictions were made. Drawing on a prior study,[41] power analyses indicated that a sample of approximately 2,800 (approximately 700 in each experimental group) would provide 90% power to detect an effect of odds ratio of 0.84 or more extreme, assuming 44.8% of participants in the control arm selected red meat. To ensure sufficient sample size after excluding vegetarians and vegans (who comprise 5% to 10% of the population[68,69]), the target sample size was approximately 3,100 respondents with complete data.

Figure 3 (available at www.jandonline.org) depicts the participant flowchart. Following the preregistered analysis plan, primary analyses excluded participants identifying as vegan or vegetarian ($n = 289$) because the intervention was not expected to affect them, yielding an analytic sample of 2,773. Two sensitivity analyses were conducted: 1 included responses from vegans and vegetarians ($n = 3,051$, preregistered) and 1 was a modified intent-to-treat analysis that included all participants who were randomized, except for those who requested removal in the debrief ($n = 3,439$; not preregistered).

Analyses of the primary outcome used logistic regression, regressing selection of red meat on indicator variables for messaging group. Separate models were used for the full-service and quick-service ordering tasks. Analyses calculated the average differential effect (ADE; ie, difference in predicted probability of selecting red meat between groups) of each red-meat–reduction message compared with the control message[70] and examined whether these effects differed from one another by comparing ADEs using χ^2 tests. Analyses also examined whether exposure to any of the 3 red-meat–reduction messages reduced selection of red meat compared with the control by collapsing across message topics. Analyses used an approach similar to that of the primary analysis to examine message impacts on calories and nutrients selected using separate linear regression models for each outcome and restaurant.

To examine whether the impact of the red-meat–reduction messages varied by demographic characteristics, moderation analyses regressed red meat selection on participant characteristics, indicator variables for messaging group, and interactions between participant characteristics and messaging group. Analyses used separate models for each characteristic and restaurant type and examined the following 8 potential moderators: age, gender, education, political affiliation, race and ethnicity, income, frequency of eating at restaurants, and frequency of red meat consumption.

To provide further insight on potential mechanisms of impact, exploratory (ie, not preregistered) analyses also examined the impact of the red-meat–reduction messages on selection of items that featured red meat as the primary ingredient. To provide insight on what participants selected in place of red meat, additional exploratory analyses used multinomial logistic regression to examine the impact of the messages on probability of selecting an item containing red meat, an item containing poultry or fish but no red meat, or a vegetarian item (ie, 3-level outcome variable).

Finally, analyses used linear regression to examine the impact of the red-meat–reduction messages on message reactions, perceptions, and intentions; these variables were standardized before analyses to facilitate comparisons across outcomes assessed on different scales. For outcomes assessed with multiple items, responses were averaged across items.

Analyses used critical $\alpha = .05$ and 2-tailed statistical tests. Analyses were conducted using Stata/MP, version 17.1[71] in 2022. De-identified data and analysis code are available upon request.

RESULTS

Sample

Mean (SD) age in the analytic sample was 46.4 (18.7) years. Approximately 12% of participants identified as Latino/a (regardless of race), and 67% identified as White, 19% as Black, and 7% as other or multiracial (regardless of ethnicity) (Table 2). The study sample was similar to the United States overall in age, race, and education distributions, but included more female participants and households with lower income than the United States overall (Table 3; available at www.jandonline.org).

Selections from the Full-Service Restaurant

In the full-service restaurant ordering task, 62.5% of participants in the control group selected a menu item with red meat (Figure 4). Compared with the control message, the health red-meat–reduction message reduced likelihood of selecting red meat by 6.0 percentage points (95% CI −11.1 to −0.8 percentage points; $P = .02$). The environmental message reduced likelihood of selecting red meat by 8.8 percentage points (95% CI −14.0 to −3.7 percentage points; $P = .001$) and the animal welfare message did not (ADE = −3.3 percentage points; 95% CI −8.4 to 1.8 percentage points; $P = .20$). Among red-meat–reduction messages, the environmental message reduced red meat selection more than the animal welfare message (P for comparison = .04); no other contrasts were significant. In analyses collapsing across red-meat–reduction topics, exposure to any red-meat–reduction message reduced likelihood of selecting red meat by 6.0 percentage points (95% CI −10.2 to −1.9 percentage points; $P = .005$). Exploratory analyses examining selection of red-meat–focused items revealed somewhat stronger effects of the red-meat–reduction messages: the health message reduced selection of red-meat–focused items by 7.7 percentage points (95% CI −12.9 to −2.4 percentage points; $P = .004$), the environmental message reduced selection by 8.6 percentage points (95% CI −13.8 to −3.3 percentage points; $P = .001$), and the animal welfare message reduced selection by 6.3 percentage points (95% CI −11.5 to −1.0 percentage points; $P = .02$). Moderation analyses did not find evidence of differences across demographic groups in the messages' effects on selection of red meat (all P for interaction > .05; Table 4; available at www.jandonline.org).

When examining calories and nutrients ordered from the full-service restaurant, participants in the animal welfare or environmental message groups selected 8.16 and 7.02 fewer milligrams of cholesterol, respectively, than the control group ($P < .05$, Table 5; available at www.jandonline.org). The health message group selected items with 49.76 fewer calories ($P = .03$) and 1.24 fewer grams of saturated fat ($P = .02$) than the control group. No other differences between groups were observed.

Exploratory analyses found that both the animal welfare (ADE = 5.7 percentage points, 95% CI 1.6 to 9.7 percentage points; $P = .006$) and environmental messages (ADE = 5.5 percentage points, 95% CI 1.5 to 9.6 percentage points; $P = .007$) increased likelihood of selecting a vegetarian item from the full-service restaurant compared with the control message (Table 6; available at www.jandonline.org). The health message increased likelihood of selecting an item containing poultry or fish (ADE = 4.7 percentage points, 95% CI 0.2 to 9.3 percentage points; $P = .04$).

Sensitivity analyses revealed a generally similar pattern of results when including vegetarians and vegans (Table 7; available at www.jandonline.org) and when conducting the modified intent-to-treat analysis (Table 8; available at www.jandonline.org).

Selections from the Quick-Service Restaurant

In the quick-service-restaurant ordering task, 39.0% of participants in the control group selected an item containing red meat (Figure 4). None of the red-meat–reduction messages reduced red meat selection compared with the control message ($P > .08$), and their effects on red meat selection did not differ from one another. Similarly, analyses collapsing across red-meat–reduction messages did not find effects of the red-meat–reduction messages on selection of red meat (ADE = −3.1 percentage points, 95% CI −7.2 to 1.1 percentage points; $P = .15$). Exploratory analyses did not find that the red-meat–reduction messages affected selection of red-meat–focused items or selection of items containing poultry or fish or vegetarian items (Table 6; available at www.jandonline.org). Moderation analyses did not find differences between groups in messages' effects on selection of red meat (all P for interaction >.05; Table 9; available at www.jandonline.org). The red-meat–reduction messages did not affect calories or nutrients selected from the quick-service restaurant (Table 5; available at www.jandonline.org). Results were similar in sensitivity analyses (Tables 7 and 8; available at www.jandonline.org).

Message Reactions, Perceptions, and Intentions

All 3 red-meat–reduction messages were perceived as more effective at discouraging red meat consumption than the control message (ADEs range, .64 to .72; all, $P < .001$; Figure 5, Table 10; available at www.jandonline.org). Likewise, all 3 red-meat–reduction messages elicited stronger emotional reactions (ADEs range, .43 to .51; all, $P < .001$), more thinking about the harms of red meat (ADES range, .67 to .72; all, $P < .001$), and more reactance (ADEs range, .44 to .67; all, $P < .001$) than the control message.

Compared with the control message, the health message led to stronger perceptions that limiting red meat intake reduces animal welfare, environmental, and health harms (ADE = .16; 95% CI, .06 to .27; P =.002) and stronger perceptions that eating red meat contributes to these harms (ADE = .13; 95% CI .03 to .23; $P = .01$). By contrast, the animal welfare and environmental messages did not affect perceptions of red meat consumption (all, $P \geq .05$).

Table 2. Participant characteristics of 2,773 US adults participating in the online randomized experiment evaluating red-meat–reduction messages[a]

	Messaging Group							
	Control (n = 702)		Animal Welfare (n = 694)		Health (n = 679)		Environmental (n = 698)	
Characteristic	**n**	**%**	**n**	**%**	**n**	**%**	**n**	**%**
Age								
18-29 y	177	25	174	25	161	24	146	21
30-44 y	180	26	166	24	163	24	195	28
45-59 y	132	19	134	19	151	22	153	22
60 y or older	213	30	220	32	204	30	204	29
Gender identity								
Woman	437	63	402	58	392	58	420	60
Man	250	36	278	40	269	40	265	38
Nonbinary or another gender	11	2	10	1	12	2	10	1
Gay, lesbian, or bisexual	90	13	84	12	89	13	74	11
Latino/a or Hispanic	78	11	80	12	73	11	104	15
Race								
White	452	65	481	70	445	66	477	69
Black or African American	142	20	116	17	127	19	125	18
American Indian or Alaska Native	14	2	9	1	12	2	15	2
Asian or Pacific Islander	36	5	35	5	40	6	31	4
Other or multiracial	55	8	48	7	49	7	47	7
Education								
High school diploma or less	222	32	220	32	200	30	188	27
Some college	184	26	189	27	169	25	197	28
College graduate or associates degree	236	34	208	30	232	34	229	33
Graduate degree	57	8	73	11	72	11	81	12
Household income, annual								
$0-$24,999	224	32	198	29	202	30	222	32
$25,000-$49,999	200	29	209	30	195	29	202	29
$50,000-$74,999	114	16	126	18	114	17	111	16
$75,000 or more	156	22	154	22	160	24	156	23

(*continued on next page*)

Table 2. Participant characteristics of 2,773 US adults participating in the online randomized experiment evaluating red-meat–reduction messages[a] (*continued*)

	Messaging Group							
	Control (n = 702)		Animal Welfare (n = 694)		Health (n = 679)		Environmental (n = 698)	
Characteristic	n	%	n	%	n	%	n	%
Income <150% federal poverty level	246	36	226	33	216	32	229	33
Political party								
Democrat	318	46	282	41	276	41	296	43
Republican	178	26	203	29	184	27	178	26
Independent	171	25	178	26	187	28	188	27
Other party	27	4	26	4	24	4	29	4
Usual red meat consumption, past 30 d								
1 time per week or less	226	32	220	32	241	35	234	34
2-3 times per week	292	42	268	39	242	36	254	36
4-6 times per week	97	14	115	17	104	15	122	17
1 time per day or more	87	12	91	13	92	14	88	13
Frequency of eating at restaurants								
Never or less than 1 time per week	285	41	282	41	281	42	280	40
1-2 times per week	305	44	280	41	267	40	283	41
3-4 times per week	90	13	102	15	100	15	104	15
5 or more times per week	18	3	26	4	24	4	26	4

[a]Missing data ranged from 0.0% to 1.4%.

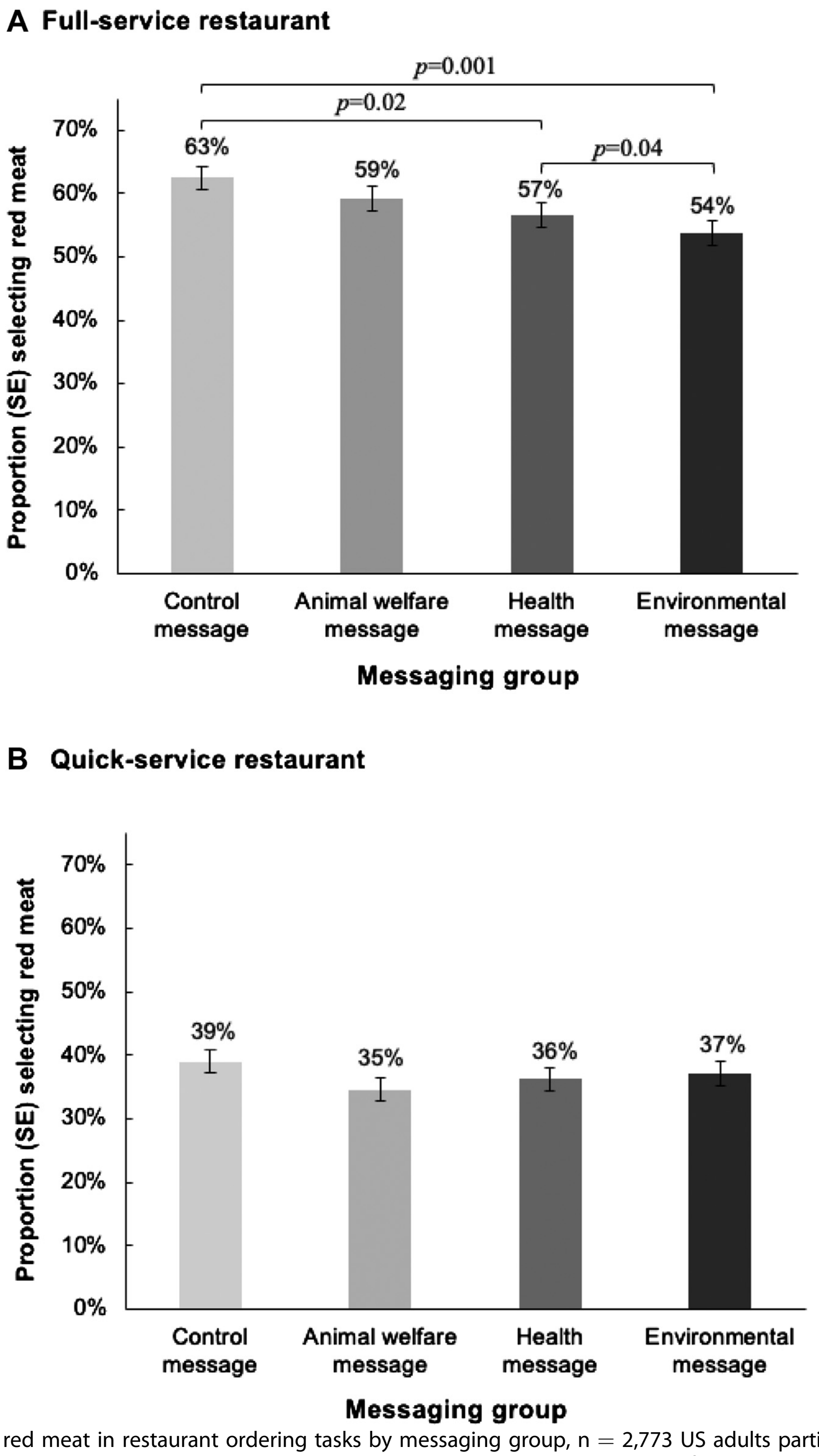

Figure 4. Selection of red meat in restaurant ordering tasks by messaging group, n = 2,773 US adults participating in the online randomized experiment evaluating red-meat–reduction messages. Figure shows proportion of participants who selected a menu item containing red meat in the restaurant ordering task at the full-service restaurant (A) and the quick-service restaurant (B). *P* values indicate differences in likelihood of selecting red meat between experimental groups; comparisons without brackets are not significantly different from one another ($P > .05$).

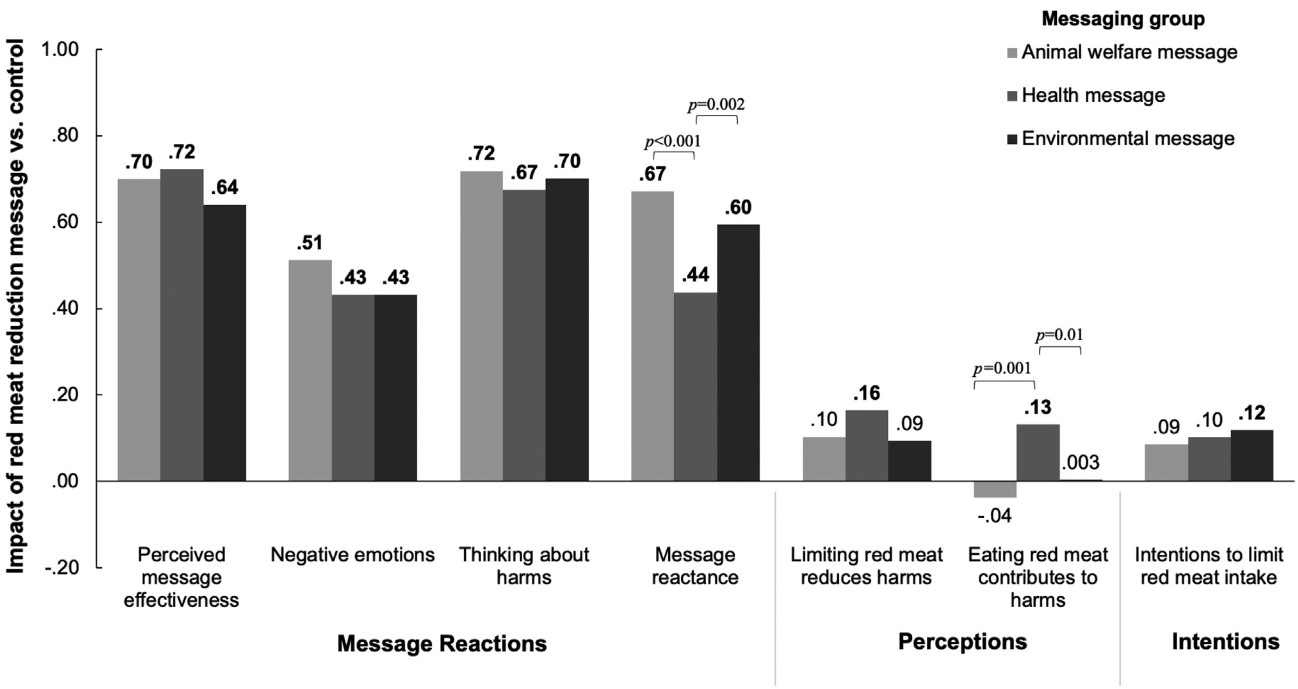

Figure 5. Impacts of red-meat-reduction messages (compared with the control message) on psychological outcomes, n = 2,773 US adults participating in the online randomized experiment evaluating red-meat−reduction messages. Figure shows average differential effects (ADEs) of each red-meat−reduction message compared with the control message, as estimated using linear regression. Dependent variables were standardized before analysis, so impacts can be interpreted as differences in SDs of the outcome. ADEs in bold type are significantly different from the control message ($P < .05$). P values indicate differences between ADEs; comparisons without brackets are not significantly different from one another ($P > .05$).

Exposure to the animal welfare, health, or environmental message led to stronger intentions to limit red meat consumption than the control message (ADEs range, .09 to .12), but this effect was only statistically significant for the environmental message ($P = .02$).

DISCUSSION

In this randomized experiment, exposure to red-meat−reduction messages focused on health or environmental (but not animal welfare) harms reduced hypothetical selection of red meat from an online full-service restaurant. The observed reductions in likelihood of ordering red meat (6 to 9 percentage points) could be meaningful at the population level, given that restaurants account for a substantial portion of red meat intake in the United States.[1,72] Moreover, all 3 red-meat−reduction messages were perceived to be more effective, elicited stronger emotional reactions, and led to more thinking about harms than the control message, reactions that are predictive of longer-term behavior change.[61,73−76] Together, these results suggest that red-meat−reduction messages hold promise for discouraging red meat selection in certain restaurant settings.

The red-meat−reduction messages reduced selection of red meat at the full-service restaurant, but not at the quick-service restaurant. One potential explanation for the divergent results by setting could be that red meat selection in the control group was higher at the full-service restaurant (62.5%) than the quick-service restaurant (39.0%), offering more room for reductions in red meat selection at the full-service restaurant. The 2 restaurants also offered a different mix of items: 40% of the full-service menu items contained red meat, compared with 26% of the quick-service restaurant. Divergent results by restaurant type could also be driven by differences in the relative price or perceived quality of red meat vs non−red meat items. For example, the non−red meat items were less expensive than the red meat items at the quick-service restaurant (mean \$8.91 and \$10.89, respectively) but were similarly priced at the full-service restaurant (\$11.17 and \$11.47, respectively); if participants believed higher-priced items were of higher quality, the lower relative price of non−red meat items at the quick-service restaurant could have provided a disincentive from selecting these options. Another possibility is that participants could more easily identify (and avoid) items containing red meat on the full-service menu than the quick-service menu, because most of the red-meat−containing full-service items prominently featured red meat (eg, burgers and steaks), and most of the quick-service items with red meat included smaller amounts of red meat as one of many ingredients (eg, turkey sandwich with bacon). Consistent with this hypothesis, exploratory analyses suggested larger effects of the red-meat−reduction messages when examining selection of red-meat−focused items specifically.

Red-meat−reduction messages focused on health or environmental harms reduced red meat selection in the full-service ordering task, and the animal welfare message did not. These results align with prior studies finding that environmentally focused labels affect meat perceptions and consumption[77−81] and that health-focused messaging interventions reduce desire for meat[46] and, among high meat consumers, meat consumption.[47] Although the animal welfare message did not reduce selection of red meat in this study, prior studies have found that interventions appealing to animal welfare can influence intentions and consumption.[34,44,45] The present study found few differences between the 3 red-meat–reduction messages, similar to most[41-43,45,46] (but not all[44,47]) prior studies comparing health, environmental, or animal welfare messages to one another.

Examination of secondary and exploratory outcomes suggests that the 3 red-meat–reduction messages could generate tradeoffs in their environmental, health, and animal welfare impacts. For example, at the full-service restaurant, the environmental message had a larger impact on red meat selection than the health message. However, only the health message reduced calories and saturated fat selected from this restaurant, suggesting potential tradeoffs between the 2 message types in environmental and health benefits. In addition, the health message simultaneously reduced selection of red meat and increased selection of poultry and fish. Although this substitution would likely reduce greenhouse gas emissions[3,82,83] and improve health outcomes,[3-5,9] it could generate greater animal suffering, given that more poultry and fish must be killed to produce the same amount of food as 1 ruminant.[84] The environmental message, by contrast, simultaneously reduced selection of red meat and increased selection of vegetarian items, suggesting a potential win-win-win for health, the environment, and animal welfare.[85]

Strengths of this study include the large national sample, randomized design, and use of realistic online restaurant menus. Limitations include that participants viewed only portions of 2 full menus and selected hypothetical meals in the context of an online experiment; generalizability to real-world restaurant choices remains to be established. In addition, social desirability bias may have influenced results, although responses were anonymous, the survey incentivized participants to select items they actually wanted, and the study's purpose was not revealed to participants.[86,87] Finally, participants were a convenience sample recruited by a survey vendor, which could increase the potential for selection bias[88]; future research will establish generalizability of these findings to other populations.

CONCLUSIONS

Messages linking red meat to health or environmental harms could lead to reductions in red meat selection at full-service restaurants.

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STATEMENT OF POTENTIAL CONFLICT OF INTEREST

No potential conflict of interest was reported by the authors.

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AUTHOR CONTRIBUTIONS

A. H. Grummon had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Study concept and design: A. H. Grummon, A. A. Musicus, A. N. Thorndike, E. B. Rimm. Acquisition, analysis, or interpretation of the data: A. H. Grummon, M. G. Salvia. Drafting of the manuscript: A. H. Grummon. Critical revision of the manuscript for important intellectual content: A. H. Grummon, A. A. Musicus, M. G. Salvia, A. N. Thorndike, E. B. Rimm. Statistical analysis: A. H. Grummon, M. G. Salvia.

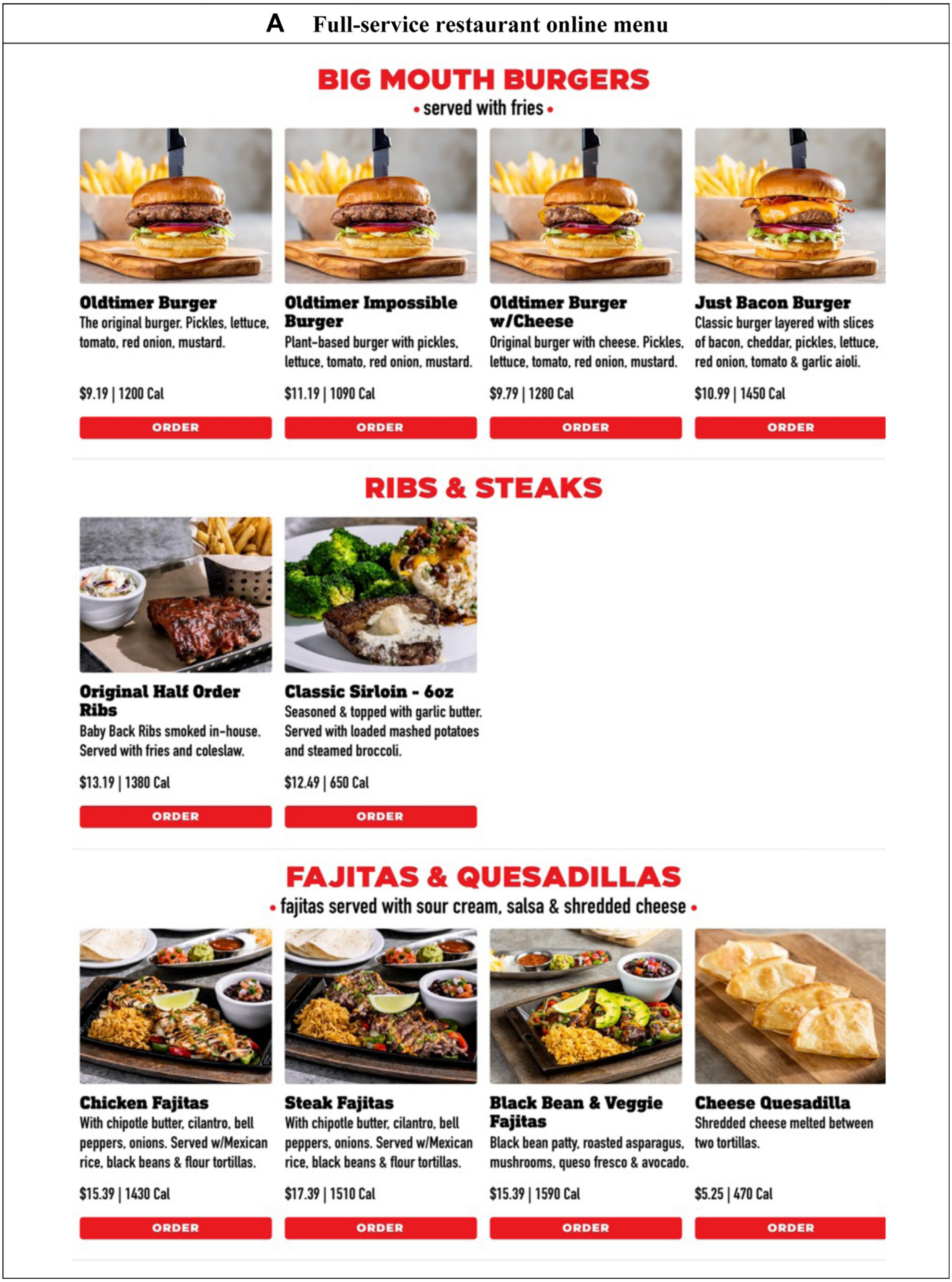

Figure 2. Online restaurant menus from a full-service restaurant (modeled after Chili's, panel A) and a quick-service restaurant (modeled after Panera Bread, panel B) used in the online randomized experiment evaluating red-meat−reduction messages.

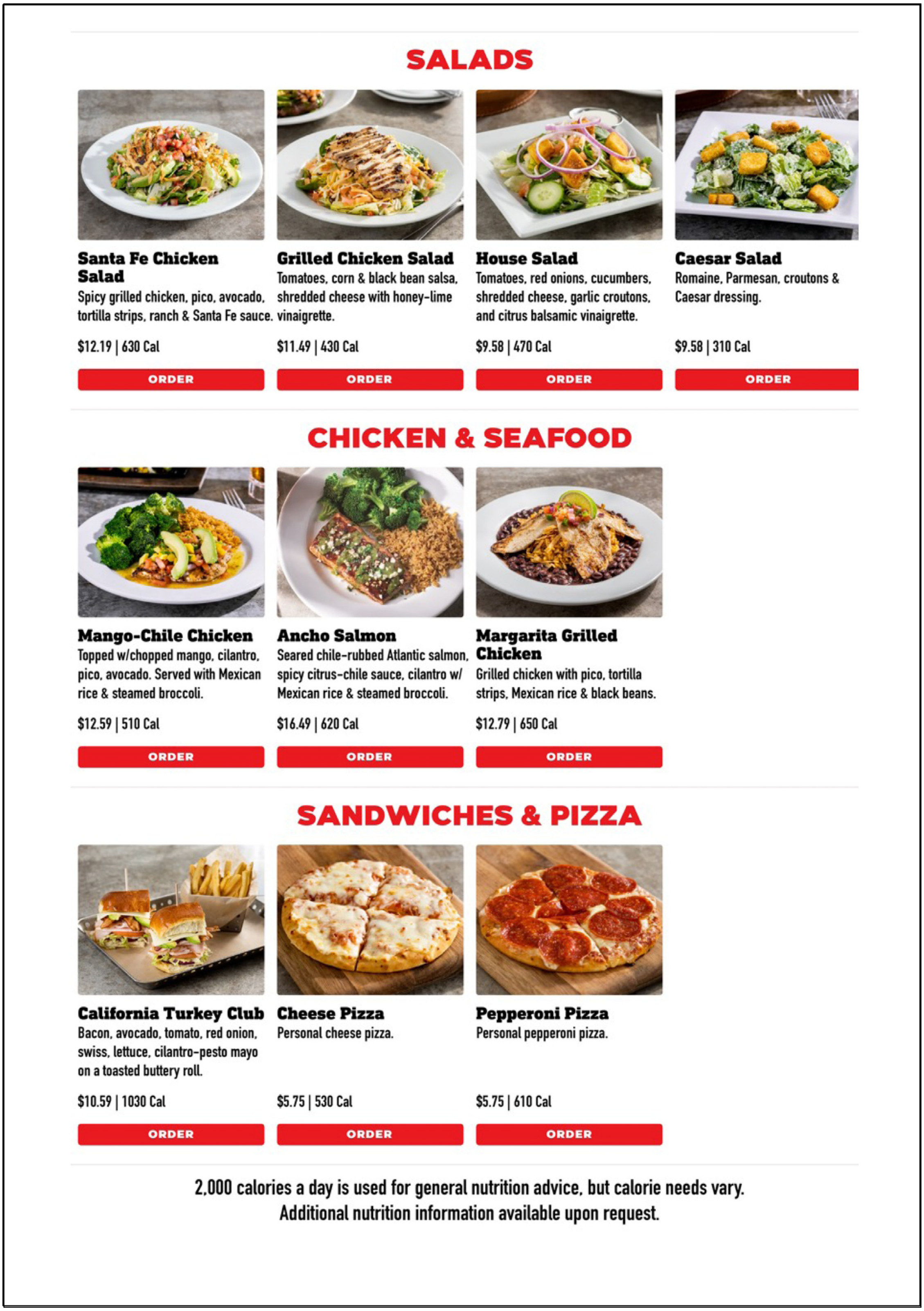

Figure 2. (*continued*) Online restaurant menus from a full-service restaurant (modeled after Chili's, panel A) and a quick-service restaurant (modeled after Panera Bread, panel B) used in the online randomized experiment evaluating red-meat—reduction messages.

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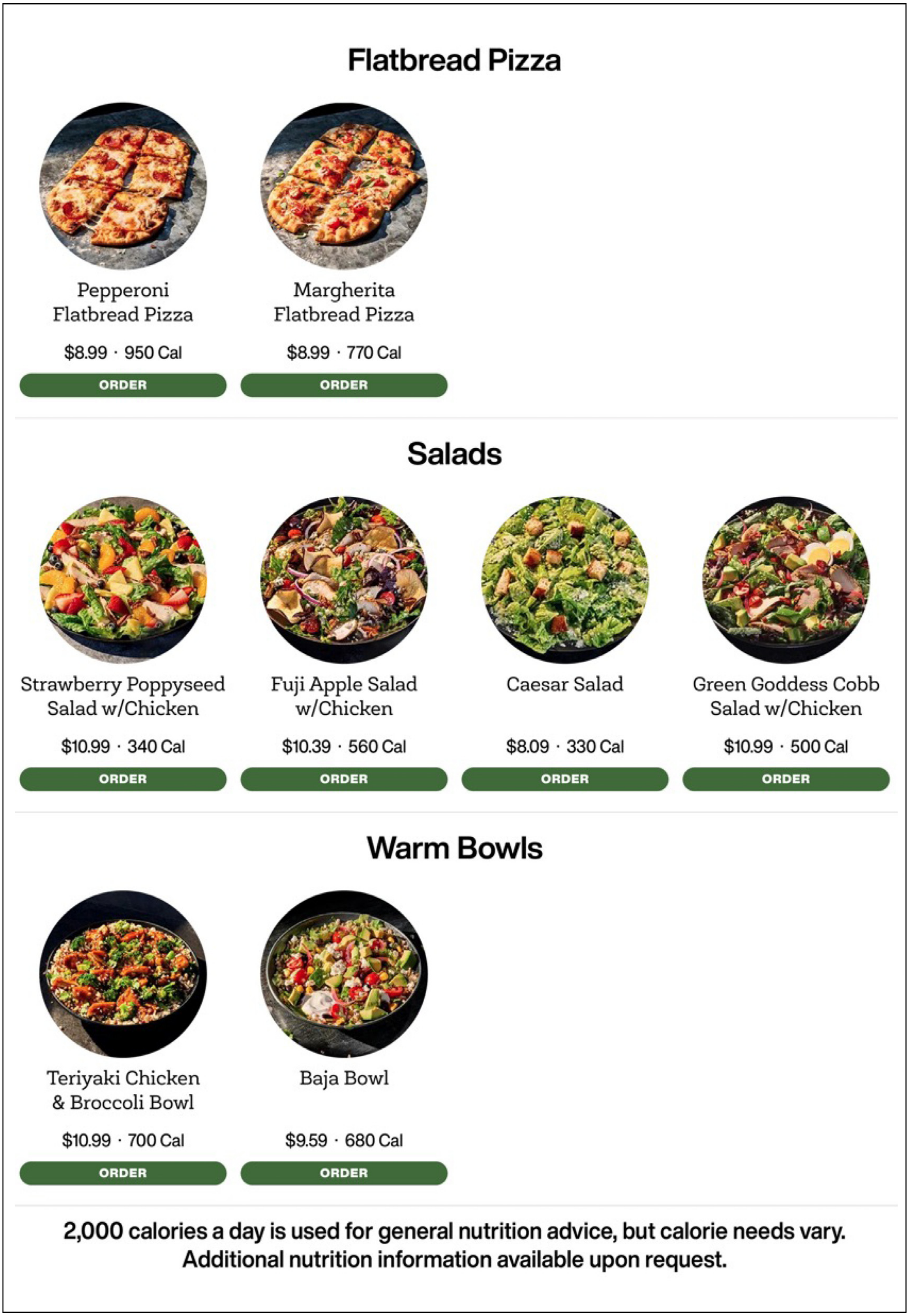

Figure 2. (*continued*) Online restaurant menus from a full-service restaurant (modeled after Chili's, panel A) and a quick-service restaurant (modeled after Panera Bread, panel B) used in the online randomized experiment evaluating red-meat−reduction messages.

Table 1. Survey measures used in the online experiment evaluating red-meat−reduction messages

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
Age	How old are you?	[Open ended numeric response]	—
Red meat consumption	The next question is about red meat. Red meat includes beef, lamb, pork, sausage, and ham. It also includes processed red meats like bacon, hot dogs, and pepperoni, and it includes products that contain red meat (eg, beef soup). It does NOT include chicken, turkey, or seafood products. In the past 30 days, how often did you eat red meat?	0 = Never 1 = Less than 1 time per week 2 = 1 time per week 3 = 2-3 times per week 4 = 4-6 times per week 5 = 1 time per day 6 = 2 times per day 7 = 3 or more times per day	—
Red meat message experiment			
Introduction and first message exposure			
Introductory prompt	The next section is about choices at restaurants. First, we will show you a message. Then we will ask you to look at menus from 2 different restaurants.	—	—
Programming notes	[randomly assign participants to an experimental group: control, animal welfare message, health message, environmental sustainability message]	—	—
Prompt and first message exposure for participants assigned to any red-meat−reduction message	[display for participants in the animal welfare message, health message, or environmental message groups] The message below is about red meat. Red meat includes beef, lamb, pork, sausage, and ham. It also includes processed red meats like bacon, hot dogs, and pepperoni, and it includes products that contain red meat (eg, beef soup). It does NOT include chicken, turkey, or seafood products. Please read the message closely. Then, continue to the next section. [display participants' assigned message]	—	—
		—	—

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Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
Prompt and first message exposure for participants assigned to control message	[display for participants in the control group] Please read the message closely. Then, continue to the next section. [display participants' assigned message]		
Programming notes	[randomize order of full-service vs quick-service restaurant ordering tasks]	—	—
	First restaurant ordering task		
Programming notes	[show quick-service vs full-service restaurant ordering task based on participants' assigned order]	—	—
Quick-service restaurant ordering task	Below is a menu from Panera. Please imagine that you are ordering from this restaurant right now. Take a few minutes to look at the choices on the menu. Then, click on the item you would like to order. We will randomly select 25 respondents to receive a bonus for completing this part of the survey. If you are selected for the bonus, we will send you either: (1) the item you selected delivered to an address of your choice or (2) an electronic gift card equal to the value of the item. You should choose the item you most want to order because it could be delivered to you. You should select one (1) item. After you have made your choice, move to the next page. [display Panera's menu]	—	—
Full-service restaurant ordering task	Below is a menu from Chili's. Please imagine that you are ordering from this restaurant right now. Take a few minutes to look at the choices on the menu. Then, click on the item you would like to order. We will randomly select 25 respondents to receive a bonus for completing this part of the survey. If you are selected for the bonus, we will send you either: (1) the item you selected delivered to an address of your choice or (2) an electronic gift card equal to the value of the item. You should choose the item you most want to order because it could be delivered to you.	—	—

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Table 1. Survey measures used in the online experiment evaluating red-meat−reduction messages (*continued*)

Construct	Item Eligibility and red meat consumption	Response scale	Scale reliability (Cronbach's α)
	You should select one (1) item. After you have made your choice, move to the next page. [display Chili's menu]		
Second message exposure			
Prompt for second message exposure	[display for all message conditions] The message below is the same message you saw previously. Please read the message again. Then, continue to the next section. [display participants' assigned message]	—	—
Second restaurant ordering task			
Programming notes	[show quick-service vs full-service restaurant ordering task based on participants' assigned order]	—	—
Quick-service restaurant ordering task	Below is a menu from Panera. Please imagine that you are ordering from this restaurant right now. Take a few minutes to look at the choices on the menu. Then, click on the item you would like to order. We will randomly select 25 respondents to receive a bonus for completing this part of the survey. If you are selected for the bonus, we will send you either: (1) the item you selected delivered to an address of your choice or (2) an electronic gift card equal to the value of the item. You should choose the item you most want to order because it could be delivered to you. You should select one (1) item. After you have made your choice, move to the next page. [display Panera's menu]	—	—
Full-service restaurant ordering task	Below is a menu from Chili's. Please imagine that you are ordering from this restaurant right now. Take a few minutes to look at the choices on the menu. Then, click on the item you would like to order. We will randomly select 25 respondents to receive a bonus for completing this part of the survey. If you are selected for the bonus,	—	—

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Table 1. Survey measures used in the online experiment evaluating red-meat−reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
	we will send you either: (1) the item you selected delivered to an address of your choice or (2) an electronic gift card equal to the value of the item. You should choose the item you most want to order because it could be delivered to you. You should select one (1) item. After you have made your choice, move to the next page. [display Chili's menu]		
Psychological reactions to messages			
Prompt	Next, we would like to ask you some questions about the message we showed you previously. The message is shown again below. Please read the message carefully. Then, answer the questions. [display participants' assigned message]	—	—
Perceived message effectiveness	How much does this message discourage you from wanting to eat red meat?	1 = Not at all 2 = Very little 3 = Somewhat 4 = Quite a bit 5 = A great deal	—
Thinking about harms	How much does this message make you think about the health harms of eating red meat?	1 = Not at all 2 = Very little 3 = Somewhat 4 = Quite a bit 5 = A great deal	.96
	How much does this message make you think about the environmental harms of producing red meat?	1 = Not at all 2 = Very little 3 = Somewhat 4 = Quite a bit 5 = A great deal	
	How much does this message make you think about how producing red meat contributes to animal suffering?	1 = Not at all 2 = Very little 3 = Somewhat	

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Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
		4 = Quite a bit 5 = A great deal	
Negative emotional reactions — question stem	[display participants' assigned message] [format as a matrix, randomize order of emotional responses] How much does this message make you feel…	1 = Not at all 2 = A little bit 3 = Somewhat 4 = Quite a bit 5 = A great deal	.94
Negative emotional reactions — worry	Worried?	—	
Negative emotional reactions — fear	Scared?	—	
Negative emotional reactions — shame	Ashamed?	—	
Negative emotional reactions — guilt	Guilty?	—	
Negative emotional reactions — regret	Regretful?	—	
Message reactance	This message is trying to manipulate me.	1 = Not at all 2 = A little bit 3 = Somewhat 4 = Quite a bit 5 = A great deal	.84
	The harm described in this message is overblown.	1 = Not at all 2 = A little bit 3 = Somewhat 4 = Quite a bit 5 = A great deal	
	This message annoys me.	1 = Not at all 2 = A little bit 3 = Somewhat	

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Table 1. Survey measures used in the online experiment evaluating red-meat–reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
		4 = Quite a bit 5 = A great deal	
Prompt	The next questions are about red meat. Red meat includes beef, lamb, pork, sausage, and ham. It also includes processed red meats like bacon, hot dogs, and pepperoni, and it includes products that contain red meat (eg, beef soup). It does not include chicken, turkey, or seafood products.	—	—
Positive perceptions of limiting red meat intake	Limiting how much red meat I eat protects my health.	1 = Not at all true 2 = Slightly true 3 = Somewhat true 4 = Very true 5 = Extremely true	.92
	Limiting how much red meat I eat is one way I can reduce my risk of health problems.	1 = Not at all true 2 = Slightly true 3 = Somewhat true 4 = Very true 5 = Extremely true	
	Limiting how much red meat I eat protects the environment.	1 = Not at all true 2 = Slightly true 3 = Somewhat true 4 = Very true 5 = Extremely true	
	Limiting how much red meat I eat is one way I can help reduce harm to the environment.	1 = Not at all true 2 = Slightly true 3 = Somewhat true 4 = Very true 5 = Extremely true	
	Limiting how much red meat I eat protects animals from suffering.	1 = Not at all true 2 = Slightly true	

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Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item Eligibility and red meat consumption	Response scale	Scale reliability (Cronbach's α)
		3 = Somewhat true 4 = Very true 5 = Extremely true	
	Limiting how much red meat I eat is one way I can help reduce harm to animals.	1 = Not at all true 2 = Slightly true 3 = Somewhat true 4 = Very true 5 = Extremely true	
Negative perceptions of eating red meat	How good or bad for your health would it be for you to eat red meat every day?	5 = Very bad 4 = Somewhat bad 3 = Neither good nor bad 2 = Somewhat good 1 = Very good	.87
	How good or bad for the environment would it be for you to eat red meat every day?	5 = Very bad 4 = Somewhat bad 3 = Neither good nor bad 2 = Somewhat good 1 = Very good	
	How good or bad for animal suffering would it be for you to eat red meat every day?	5 = Very bad 4 = Somewhat bad 3 = Neither good nor bad 2 = Somewhat good 1 = Very good	
Prompt	The next statements are about the next week (7 days).	—	.96
Intentions to limit red meat intake	In the next week, I want to limit how much red meat I eat.	1 = Not at all 2 = A little bit 3 = Somewhat 4 = Quite a bit 5 = Very much	

(*continued on next page*)

Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
	In the next week, I plan to limit how much red meat I eat.	1 = Not at all 2 = A little bit 3 = Somewhat 4 = Quite a bit 5 = Very much	
	In the next week, I am likely to limit how much red meat I eat.	1 = Not at all likely 2 = A little bit likely 3 = Somewhat likely 4 = Quite likely 5 = Very likely	
Demographics			
Prompt	We are asking the questions in the next section to better understand who completed our survey.	—	—
Gender	How do you identify?	1 = Woman 2 = Man 3 = Non-binary 4 = Prefer to self-describe	—
Transgender identity	Do you identify as transgender or gender nonconforming?	1 = Yes 0 = No 2 = Prefer not to say	—
Hispanic/Latino ethnicity	Are you of Hispanic, Latino, or Spanish origin?	1 = Yes 0 = No	—
Race	What is your race? (Check all that apply).	[Check all that apply] 1 = White 2 = Black or African American 3 = American Indian or Alaska Native 4 = Asian 5 = Native Hawaiian or Other Pacific Islander 6 = Some other race:	—

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Table 1. Survey measures used in the online experiment evaluating red-meat−reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
Education	What is the highest degree or level of school you have completed?	1 = Less than high school 2 = High school graduate (or General Educational Development test) 3 = Some college or technical school 4 = Associate's degree 5 = Bachelor's degree 6 = Graduate or professional degree	—
Vegetarian/vegan	Are you vegetarian or vegan?	0 = No 1 = Yes	—
Frequency of eating at restaurants	In a typical week, how often do you go out to eat or buy ready-to-eat foods from restaurants? This includes fast food restaurants, fast causal restaurants, and sit-down restaurants. It also includes delivery or takeout orders.	1 = Never or less than 1 time per week 2 = 1 to 2 times per week 3 = 3 to 4 times per week 4 = 5 or more times per week	—
Sexual orientation	The next question is about your sexual orientation. Do you consider yourself to be…	1 = Straight or heterosexual 2 = Gay or lesbian 3 = Bisexual 4 = Prefer to self-describe:	—
Party identification	Do you consider yourself a Democrat, a Republican, an Independent, or something else?	1 = Strong Democrat 2 = Moderate Democrat 3 = Lean Democrat 4 = Independent 5 = Lean Republican 6 = Moderate Republican 7 = Strong Republican 8 = Something else:	—
Household size	How many people are in your household, including you?	No. of people [restricted to 1 to 20, whole numbers]	—
No. of children	How many children (aged 0 through 18 y) currently live in your household?	[free response, numeric entry restricted to 0-15, whole numbers]	—

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Table 1. Survey measures used in the online experiment evaluating red-meat–reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
Income	Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$10,000 2 = \$10,000-\$14,999 3 = \$15,000-\$24,999 4 = \$25,000-\$34,999 5 = \$35,000-\$49,999 6 = \$50,000-\$74,999 7 = \$75,000-\$99,999 8 = \$100,000-\$149,999 9 = \$150,000-\$199,999 10 = \$200,000 or more	—
Income to poverty ratio	[ask only if household size = 1] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$19,140 2 = Between \$19,140 and \$25,519 3 = Between \$25,520 and \$31,899 4 = Between \$31,900 and \$38,279 5 = \$38,280 or more	—
	[ask only if household size = 2] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$25,860 2 = Between \$25,860 and \$34,479 3 = Between \$34,480 and \$43,099 4 = Between \$43,100 and \$51,719 5 = \$51,720 or more	—
	[ask only if household size = 3] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$32,580 2 = Between \$32,580 and \$43,439 3 = Between \$43,440 and \$54,299 4 = Between \$54,300 and \$65,159 5 = \$65,160 or more	—
	[ask only if household size = 4] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$39,300 2 = Between \$39,300 and \$52,399 3 = Between \$52,400 and \$65,499 4 = Between \$65,500 and \$78,599 5 = \$78,600 or more	—

(*continued on next page*)

Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
	[Ask only if household size = 5] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $46,020 2 = Between $46,020 and $61,359 3 = Between $61,360 and $76,699 4 = Between $76,700 and $92,039 5 = $92,040 or more	—
	[ask only if household size = 6] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $52,740 2 = Between $52,740 and $70,319 3 = Between $70,320 and $87,899 4 = Between $87,900 and $105,479 5 = $105,480 or more	—
	[ask only if household size = 7] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $59,460 2 = Between $59,460 and $79,279 3 = Between $79,280 and $99,099 4 = Between $99,100 and $118,919 5 = $118,920 or more	—
	[ask only if household size = 8] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $66,180 2 = Between $66,180 and $88,239 3 = Between $88,240 and $110,299 4 = Between $110,300 and $132,359 5 = $132,360 or more	—
	[ask only if household size = 9] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $72,900 2 = Between $72,900 and $97,199 3 = Between $97,200 and $121,499 4 = Between $121,500 and $145,799 5 = $145,800 or more	—
	[ask only if household size = 10] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $79,620 2 = Between $79,620 and $106,159 3 = Between $106,160 and $132,699 4 = Between $132,700 and $159,239 5 = $159,240 or more	—

(*continued on next page*)

Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
	[ask only if household size = 11] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $86,340 2 = Between $86,340 and $115,119 3 = Between $115,120 and $143,899 4 = Between $143,900 and $172,679 5 = $172,680 or more	—
	[ask only if household size = 12] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $93,060 2 = Between $93,060 and $124,079 3 = Between $124,080 and $155,099 4 = Between $155,100 and $186,119 5 = $186,120 or more	—
	[Ask only if household size = 13] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $99,780 2 = Between $99,780 and $133,039 3 = Between $133,040 and $166,299 4 = Between $166,300 and $199,559 5 = $199,560 or more	—
	[Ask only if household size = 14] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $106,500 2 = Between $106,500 and $141,999 3 = Between $142,000 and $177,499 4 = Between $177,500 and $212,999 5 = $213,000 or more	—
	[Ask only if household size = 15] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $113,220 2 = Between $113,220 and $150,959 3 = Between $150,960 and $188,699 4 = Between $188,700 and $226,439 5 = $226,440 or more	—
	[Ask only if household size = 16] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than $119,940 2 = Between $119,940 and $159,919 3 = Between $159,920 and $199,899 4 = Between $199,900 and $239,879 5 = $239,880 or more	—

(*continued on next page*)

Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
	[Ask only if household size = 17] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$126,660 2 = Between \$126,660 and \$168,879 3 = Between \$168,880 and \$211,099 4 = Between \$211,100 and \$253,319 5 = \$253,320 or more	—
	[Ask only if household size = 18] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$133,380 2 = Between \$133,380 and \$177,839 3 = Between \$177,840 and \$222,299 4 = Between \$222,300 and \$266,759 5 = \$266,760 or more	—
	[Ask only if household size = 19] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$140,100 2 = Between \$140,100 and \$186,799 3 = Between \$186,800 and \$233,499 4 = Between \$233,500 and \$280,199 5 = \$280,200 or more	—
	[Ask only if household size = 20] Which of the following categories best describes your total household income in the last 12 months?	1 = Less than \$146,820 2 = Between \$146,820 and \$195,759 3 = Between \$195,760 and \$244,699 4 = Between \$244,700 and \$293,639 5 = \$293,640 or more	—
Debrief and closure			
Debrief text	Thank you for taking part in this research study. There was some information about the study that we did not share with you at the beginning of your participation, so that you would react honestly. We would now like to fully inform you about the nature of this research and explain a bit more about the survey you just completed. In the survey we indicated that 25 participants would either receive one of the foods/beverages they selected from the sets of packaged food and beverage products shown in the survey or the value of the	—	—

(*continued on next page*)

Table 1. Survey measures used in the online experiment evaluating red-meat—reduction messages (*continued*)

Construct	Item	Response scale	Scale reliability (Cronbach's α)
	Eligibility and red meat consumption		
	item as an electronic gift card. In actuality, all participants selected for this bonus will receive a $10 gift card (slightly more than the value of the most expensive item we showed you in the survey) in addition to the amount agreed upon when you entered this survey. We stated that you might receive the meal and/or product you selected because we wanted you to behave as you normally would when choosing foods and to choose options you would actually want to purchase. This study was conducted by researchers at the [INSTITUTION BLINDED] who are interested in the effects of messages about red meat and about different front-of-package labels for foods and beverages. We will use this study to determine how effective different types of messages and front-of-package labels are at changing which products people choose. Please feel free to contact the investigator, [BLINDED], with any questions or if you have any concerns about your participation. She can be reached via email at [BLINDED]. If you would like a copy of this debriefing form, please save this page to your computer.		
Request to remove data	Please choose one of the options below.	1 = I agree to have my data included in this study. 2 = I no longer wish to participate and/or I would like my data removed from the study. (Your data will be removed from the study entirely, except for the consent and debriefing form).	—
Confirm request	[display if option 2 selected in previous question] Please confirm that you would like your data removed from the study.	1 = Confirm, remove my data from the study 2 = Keep my data in the study	—
Closure	You have now finished this survey. Thank you for your participation!	—	—

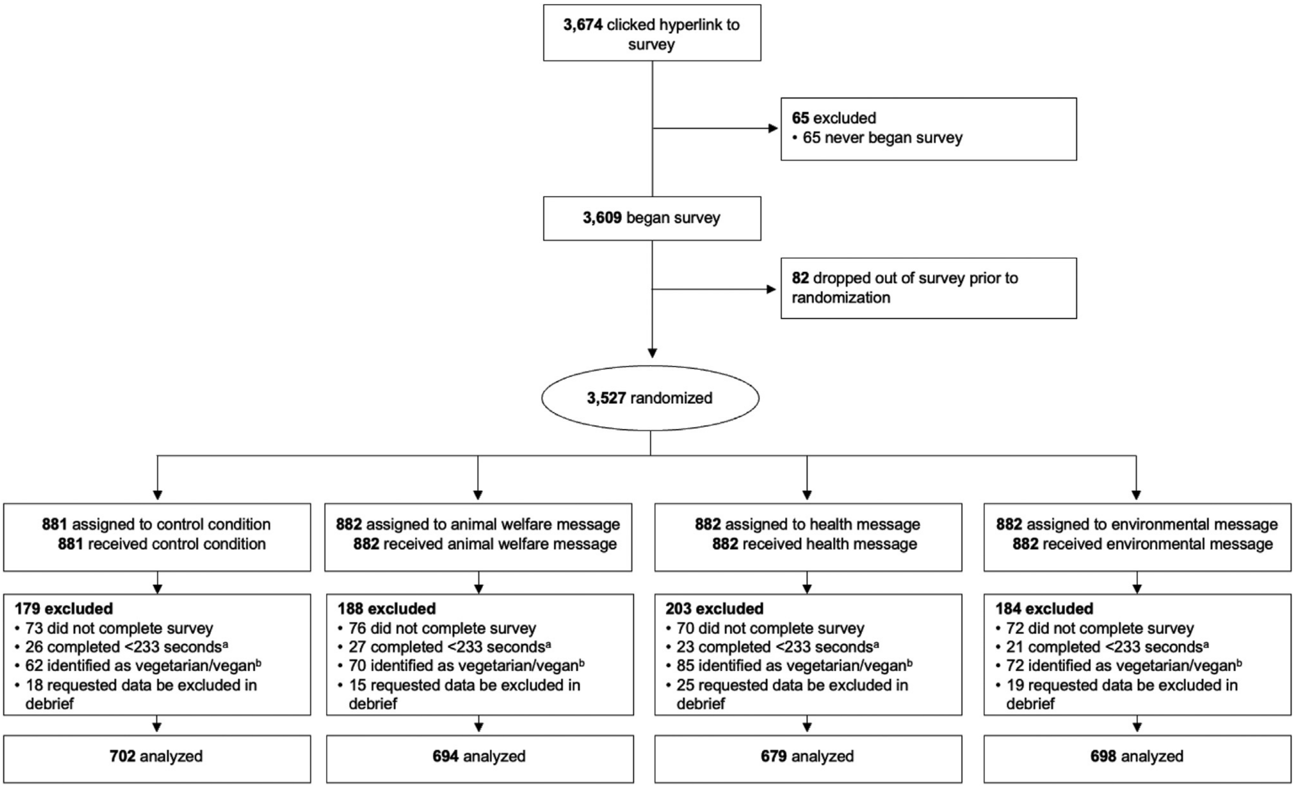

Figure 3. Consolidated Standards of Reporting Trials flow diagram for the online randomized experiment evaluating red-meat—reduction messages.

[a]One-third of median completion time in a soft launch.

[b]Vegetarians and vegans were excluded in primary analyses because the intervention was not expected to affect their behavior. Sensitivity analyses included vegetarians and vegans.

Table 3. Comparison of key characteristics of the study sample of 3,051 US adults participating in the online randomized experiment evaluating red-meat—reduction messages) with national estimates[a]

Characteristic	Study sample	National estimate
	←——— % ———	———→
Age		
18-29 y	25.4	21.0
30-44 y	25.8	25.2
45-59 y	20.0	24.4
60 y or older	28.8	29.4
Gender/sex[b]		
Female	61.3	51.3
Male	38.7	48.7
Gay, lesbian, or bisexual[c]	13.0	5.6
Latino/a or Hispanic	12.7	16.4
Race		
White	66.1	73.6
Black or African American	18.9	12.5
American Indian or Alaska Native	1.9	0.8
Asian or Pacific Islander	5.5	6.1
Other or multiracial	7.6	7.0
Education		
High school diploma or less	29.9	39.0
Some college	26.2	22.1
College graduate or associates degree	33.4	27.5
Graduate degree	10.5	11.4
Household income, annual[d]		
$0-$24,999	31.0	18.1
$25,000-$49,999	28.6	19.7
$50,000-$74,999	17.1	16.5
$75,000 or more	23.3	45.8

[a]To maximize comparability with national estimates, the table presents statistics for the full sample, including vegans and vegetarians. National estimates for age, gender/sex, ethnicity, race, and education are survey-weighted estimates among adults (18 years and older) in the 2019 American Community Survey (ACS) 1-Year Public Use Microdata Sample (PUMS).[89] The 2019 ACS data were used because the Census Bureau did not release its standard 2020 ACS 1-year PUMS data because of the impacts of the COVID-19 pandemic and the 2021 data were not yet available.

[b]Estimates for gender/sex for the study sample exclude participants who identified as nonbinary or another gender because the ACS only includes information on identification as male or female.

[c]The national estimate for identification as gay, lesbian, or bisexual is from a national Gallup Poll conducted in 2020.[90]

[d]National estimates for income are from the Current Population Survey, 2020, the most recently available year of data.[91]

Table 4. Impacts of red-meat–reduction messages vs control on selection of red meat in full-service restaurant ordering task, by demographic group, of 2,773 US adults participating in an online randomized experiment evaluating red-meat–reduction messages

	Impact of Red-Meat–Reduction Message (vs Control) on Likelihood of Selecting Red Meat			
Characteristic	**Animal welfare message**	**Health message**	**Environmental message**	***P* value[a]**
	←—%, *ADE*[b] *(95% CI)*—→			
Age				.46
18-29 y	−7.9 (−18.3 to 2.5)	−13.0 (−23.8 to −2.3)	−3.5 (−13.6 to 6.6)	
30-44 y	1.0 (−9.3 to 11.2)	−5.8 (−15.7 to 4.1)	5.2 (−4.8 to 15.3)	
45-59 y	−8.8 (−20.1 to 2.6)	−4.9 (−16.2 to 6.4)	−11.4 (−23.2 to 0.3)	
60 y and older	−8.1 (−17.6 to 1.4)	−11.6 (−21.1 to −2.1)	−4.6 (−13.9 to 4.7)	
Educational attainment				.39
High school/GED[c] or less	−10.2 (−19.3 to −1.2)	−8.5 (−17.6 to 0.7)	−7.5 (−16.3 to 1.2)	
Some college	−2.8 (−13.0 to 7.4)	−7.1 (−17.0 to 2.7)	−4.8 (−14.8 to 5.1)	
2–y or 4–y college degree	−7.6 (−16.6 to 1.4)	−7.4 (−16.4 to 1.6)	2.6 (−6.6 to 11.7)	
Graduate degree	7.8 (−9.5 to 25.1)	−9.6 (−26.4 to 7.2)	−2.5 (−19.8 to 14.8)	
Race and ethnicity				.49
Non-Hispanic White	−4.3 (−11.0 to 2.5)	−8.0 (−14.7 to −1.3)	−1.9 (−8.5 to 4.6)	
Non-Hispanic Black	−16.6 (−28.8 to −4.4)	−9.7 (−21.8 to 2.4)	−1.2 (−13.5 to 11.0)	
Latino/a	−7.7 (−22.9 to 7.6)	−14.1 (−28.2 to 0.007)	−10.4 (−25.4 to 4.5)	
Non-Hispanic other or multiracial	3.4 (−12.2 to 19.1)	−4.5 (−21.6 to 12.5)	−6.6 (−23.3 to 10.1)	
Political affiliation				.85
Democrat	−5.0 (−13.0 to 3.0)	−7.2 (−15.1 to 0.7)	.6 (−7.3 to 8.5)	
Republican	−9.8 (−19.7 to 0.005)	−11.8 (−21.8 to −1.8)	−6.0 (−15.5 to 3.6)	
Independent	−6.8 (−16.7 to 3.2)	−9.8 (−19.8 to 0.2)	−9.9 (−20.1 to 0.2)	
Another party affiliation	2.3 (−25.1 to 29.7)	−3.6 (−29.8 to 22.6)	9.7 (−16.9 to 36.2)	
Frequency of eating at restaurants				.73
Never or <1 time per week	−.9 (−9.1 to 7.2)	−7.5 (−15.7 to 0.7)	−1.1 (−9.3 to 7.0)	
1-2 times per week	−11.1 (−19.2 to −3.1)	−9.2 (−17.1 to −1.2)	−5.0 (−12.9 to 2.9)	
3-4 times per week	−2.7 (−16.2 to 10.9)	−9.0 (−22.6 to 4.6)	−3.9 (−17.4 to 9.6)	
5 times per week or more	−27.8 (−55.5 to −0.0)	−20.1 (−47.1 to 6.9)	−12.4 (−38.9 to 14.1)	
Frequency of eating red meat				.35
1 time per week or less	−11.2 (−20.2 to −2.2)	−8.2 (−17.3 to 0.9)	−5.8 (−15.1 to 3.4)	
2-3 times per week	−7.2 (−15.5 to 1.2)	−8.2 (−16.5 to 0.005)	−1.8 (−9.8 to 6.2)	
4-6 times per week	−4.0 (−16.4 to 8.3)	−12.8 (−25.0 to −0.5)	−6.4 (−18.6 to 5.8)	
1 time per day or more	9.1 (−3.2 to 21.4)	−8.8 (−22.5 to 5.0)	−1.0 (−14.2 to 12.2)	
Gender				.87
Female	−7.0 (−15.2 to 1.3)	−10.2 (−18.5 to −1.9)	−2.5 (−10.5 to 5.6)	
Male	−6.4 (−13.1 to 0.4)	−7.6 (−14.2 to −0.9)	−4.0 (−10.6 to 2.7)	
Household poverty level				.47
≥150% federal poverty level	−7.3 (−13.7 to −0.9)	−11.5 (−17.9 to −5.1)	−5.6 (−11.9 to 0.8)	
<150% federal poverty level	−4.3 (−13.3 to 4.6)	−3.7 (−12.5 to 5.1)	1.5 (−7.2 to 10.2)	

[a]*P* values are from Wald tests of significance of the coefficients on all interaction terms.
[b]ADE = average differential effect.
[c]GED = General Education Development test.

Table 5. Impact of red-meat–reduction messages vs control on calories and nutrients selected in restaurant ordering tasks, 2,773 US adults participating in the online randomized experiment evaluating red-meat–reduction messages[a]

Outcome	Impacts of Red Meat–Reduction Messages vs Control: Animal welfare message	Health message	Environmental message
	←ADE[b] (95% CI)→		
Full-service restaurant			
Calories, kcal	−29.40 (−73.00 to 14.19)	**−49.76 (−93.60 to −5.92)**	−15.21 (−58.74 to 28.33)
Saturated fat, g	−0.29 (−1.31 to 0.73)	**−1.24 (−2.28 to −0.21)**	−0.86 (−1.90 to 0.17)
Sodium, mg	−14.78 (−71.76 to 42.20)	−25.87 (−83.56 to 31.81)	−33.52 (−91.16 to 24.11)
Sugar, g	−0.10 (−0.69 to 0.49)	−0.20 (−0.80 to 0.40)	−0.02 (−0.61 to 0.58)
Cholesterol, mg	**−8.16 (−14.78 to −1.53)**	−4.32 (−11.02 to 2.39)	**−7.02 (−13.72 to −0.32)**
Quick-service restaurant			
Calories, kcal	−16.31 (−40.33 to 7.72)	−19.15 (−43.30 to 5.01)	−17.35 (−41.34 to 6.63)
Saturated fat, g	−0.19 (−1.14 to 0.76)	−0.57 (−1.53 to 0.39)	−0.59 (−1.54 to 0.36)
Sodium, mg	−9.90 (−70.64 to 50.83)	−26.30 (−87.37 to 34.77)	−17.50 (−78.14 to 43.15)
Sugar, g	0.22 (−0.45 to 0.89)	0.15 (−0.52 to 0.82)	−0.20 (−0.87 to 0.47)
Cholesterol, mg	−.92 (−6.46 to 4.62)	−2.71 (−8.28 to 2.86)	−5.06 (−10.59 to 0.47)

[a]Bold type indicates a statistically significant impact of the red-meat–reduction messages compared with the control condition ($P < .05$). Within each row, ADEs did not differ from one another (all p values for comparisons of ADEs were >.05).
[b]ADE = average differential effect as estimated from logistic (or binary outcomes) or linear (for continuous outcomes) regressions.

Table 6. Impact of red-meat–reduction messages vs control on selection of items containing red meat, items containing poultry or fish, or vegetarian items in restaurant ordering tasks, 2,773 US adults participating in the online randomized experiment evaluating red-meat–reduction messages[a]

Messaging group (control: reference)	Impact of Red-Meat–Reduction Messages vs Control on Probability of Selecting Different Meal Types: Item containing red meat	Item containing poultry or fish (no red meat)	Vegetarian item
	←%, ADE[b] (95% CI)→		
Full-service restaurant			
Animal welfare	−3.3 (−8.4 to 1.8)	−2.3 (−6.6 to 1.9)	**5.7 (1.6 to 9.7)**
Health	**−6.0 (−11.2 to −0.8)**	**4.7 (0.2 to 9.3)**	1.3 (−2.6 to 5.1)
Environmental	**−8.8 (−14.0 to −3.7)**	3.3 (−1.2 to 7.7)	**5.5 (1.5 to 9.6)**
Quick-service restaurant			
Animal welfare	−4.4 (−9.5 to 0.6)	2.0 (−3.0 to 7.0)	2.5 (−2.3 to 7.2)
Health	−2.8 (−7.9 to 2.3)	3.1 (−2.0 to 8.1)	−0.3 (−4.9 to 4.4)
Environmental	−1.9 (−7.0 to 3.2)	0.9 (−4.1 to 5.9)	1.0 (−3.7 to 5.7)

[a]Before fitting final models, analyses examined the independence of irrelevant alternatives (IIA) assumption using Hausman-McFadden tests. For both restaurants, analyses failed to reject the null hypothesis that the differences in coefficients are not systematic between models with different alternatives, providing evidence that IIA was not violated. Bold type indicates a statistically significant impact of the red-meat–reduction messages compared with the control condition ($P < .05$).
[b]ADE = average differential effect as estimated from multinomial logistic regression.

Table 7. Impacts of red-meat—reduction messages on selections in restaurant ordering tasks, sensitivity analyses including participants who identified as vegetarian or vegan, 3,051 US adults participating in the online randomized experiment evaluating red-meat—reduction messages[a]

Outcome	Impacts of Red-Meat—Reduction Messages vs Control		
	Animal welfare message	Health message	Environmental message
	← ADE[b] (95% CI) →		
Full-service restaurant			
Selection of red meat, %	−4.5 (−9.5 to 0.4)	**−6.3 (−11.2 to −1.3)**	**−7.5 (−12.4 to −2.6)**
Calories, kcal	−41.18 (−82.96 to 0.60)	**−48.20 (−90.00 to −6.41)**	−19.68 (−61.36 to 22.01)
Saturated fat, g	−0.46 (−1.43 to 0.51)	**−1.21 (−2.19 to −0.24)**	−0.72 (−1.69 to 0.26)
Sodium, mg	−22.03 (−77.29 to 33.22)	−19.83 (−75.47 to 35.81)	−29.25 (−85.00 to 26.49)
Sugar, g	−0.26 (−0.82 to 0.30)	−0.29 (−0.85 to 0.28)	−0.11 (−0.67 to 0.46)
Cholesterol, mg	**−8.75 (−15.14 to −2.37)**	−5.57 (−12.00 to 0.87)	**−7.04 (−13.48 to −0.60)**
Quick-service restaurant			
Selection of red meat, %	**−5.1 (−9.9 to −0.3)**	−4.2 (−9.1 to 0.6)	−2.1 (−7.0 to 2.7)
Calories, kcal	−15.84 (−38.98 to 7.30)	**−24.51 (−47.66 to −1.36)**	−16.07 (−39.16 to 7.02)
Saturated fat, g	−0.12 (−1.03 to 0.80)	−0.81 (−1.73 to 0.10)	−0.53 (−1.45 to 0.38)
Sodium, mg	−9.78 (−67.67 to 48.10)	−36.46 (−94.37 to 21.44)	−16.48 (−74.23 to 41.27)
Sugar, g	0.29 (−0.34 to 0.93)	0.09 (−0.54 to 0.72)	−0.17 (−0.80 to 0.46)
Cholesterol, mg	−1.24 (−6.65 to 4.18)	−4.71 (−10.13 to 0.70)	−5.17 (−10.58 to 0.23)

[a]Bold indicates a statistically significant impact of the red-meat—reduction messages compared with the control condition ($P < .05$). Within each row, ADEs did not differ from one another (all P values for comparisons of ADEs $> .05$).
[b]ADE = average differential effect.

Table 8. Impacts of red-meat—reduction messages on selections in restaurant ordering tasks, modified intent-to-treat sensitivity analyses, including all participants randomized except for those who requested that their data be excluded in the debrief, 3,439 US adults participating in the online randomized experiment evaluating red-meat—reduction messages[a]

	Impacts of Red-Meat—Reduction Messages vs Control		
Outcome	Animal welfare message	Health message	Environmental message
	←—— *ADE[b] (95% CI)* ——→		
Full-service restaurant (n = 3,306 with complete data)			
Selection of red meat, %	**−5.1 (−9.9 to −0.4)**	**−5.8 (−10.6 to −1.1)**	**−7.3 (−12.0 to −2.6)**
Calories, kcal	**−51.87 (−91.90 to −11.84)**	**−57.23 (−97.41 to −17.04)**	−30.63 (−70.66 to 9.40)
Saturated fat, g	−0.65 (−1.58 to 0.28)	**−1.27 (−2.20 to −0.33)**	−0.85 (−1.79 to 0.09)
Sodium, mg	−39.81 (−92.75 to 13.13)	−38.92 (−92.38 to 14.54)	−45.01 (−98.45 to 8.43)
Sugar, g	−0.42 (−0.95 to 0.12)	−0.45 (−0.99 to 0.09)	−0.23 (−0.76 to 0.31)
Cholesterol, mg	**−9.58 (−15.68 to −3.48)**	−6.07 (−12.23 to 0.09)	**−7.82 (−13.98 to −1.67)**
Quick-service restaurant (n = 3,309 with complete data)			
Selection of red meat, %	**−5.9 (−10.5 to −1.3)**	−4.4 (−9.0 to 0.3)	−3.4 (−8.1 to 1.2)
Calories, kcal	−17.19 (−39.38 to 5.00)	−23.98 (−46.21 to −1.76)	−14.48 (−36.63 to 7.66)
Saturated fat, g	−0.13 (−1.01 to 0.076)	−0.85 (−1.74 to 0.03)	−0.39 (−1.27 to 0.50)
Sodium, mg	−14.38 (−69.99 to 41.24)	−37.75 (−93.45 to 17.94)	−14.22 (−69.71 to 41.28)
Sugar, g	0.29 (−0.32 to 0.89)	0.08 (−0.52 to 0.68)	−0.21 (−0.81 to 0.39)
Cholesterol, mg	−2.24 (−7.43 to 2.96)	−5.82 (−11.02 to −0.62)	**−5.90 (−11.08 to −0.72)**

[a]Bold type indicates a statistically significant impact of the red-meat—reduction messages compared with the control condition ($P < .05$). Within each row, ADEs did not differ from one another (all P values for comparisons of ADEs $> .05$).
[b]ADE = average differential effect as estimated from logistic (or binary outcomes) or linear (for continuous outcomes) regressions.

Table 9. Impacts of red-meat—reduction messages vs control on selection of red meat in quick-service restaurant ordering task, by demographic group, 2,773 US adults participating in the online randomized experiment evaluating red-meat—reduction messages

Characteristic	Impact of Red-Meat—Reduction Message (vs Control) on Likelihood of Selecting Red Meat			*P* value[a]
	Animal welfare message	Health message	Environmental message	
	←——%, *ADE*[b] *(95% CI)*——→			
Age				.75
18-29 y	−1.0 (−11.0 to 9.1)	−3.1 (−13.3 to 7.2)	−5.7 (−15.4 to 3.9)	
30-44 y	2.0 (−8.3 to 12.3)	3.8 (−6.1 to 13.7)	−1.7 (−11.8 to 8.5)	
45-59 y	−13.8 (−25.3 to −2.3)	−10.3 (−21.9 to 1.2)	−9.7 (−21.7 to 2.2)	
60 y and older	−1.8 (−11.0 to 7.4)	−2.3 (−11.5 to 6.9)	−2.5 (−11.6 to 6.5)	
Educational attainment				.71
High school/GED[c] or less	4.8 (−4.4 to 14.0)	4.1 (−5.2 to 13.5)	−1.0 (−9.9 to 7.8)	
Some college	−7.4 (−17.6 to 2.7)	−2.3 (−12.2 to 7.6)	−7.5 (−17.4 to 2.4)	
2—y or 4-y college degree	−7.9 (−16.6 to 0.8)	−7.0 (−15.8 to 1.8)	−5.9 (−15.0 to 3.1)	
Graduate degree	−0.4 (−16.9 to 16.2)	−3.0 (−19.0 to 13.0)	−2.2 (−18.6 to 14.2)	
Race and ethnicity				.80
Non-Hispanic White	−0.7 (−7.3 to 6.0)	−1.1 (−7.7 to 5.5)	−2.6 (−9.1 to 3.9)	
Non-Hispanic Black	−8.7 (−20.5 to 3.1)	−0.7 (−12.8 to 11.4)	−12.6 (−24.5 to −0.8)	
Latino/a	−5.5 (−20.9 to 9.9)	−8.0 (−22.1 to 6.1)	−3.5 (−18.6 to 11.6)	
Non-Hispanic other or multiracial	−7.5 (−23.2 to 8.2)	−1.7 (−18.7 to 15.4)	−5.6 (−22.0 to 10.9)	
Political affiliation				.06
Democrat	−0.1 (−8.0 to 7.8)	−5.8 (−13.4 to 1.8)	−9.8 (−17.4 to −2.2)	
Republican	−9.5 (−19.3 to 0.4)	5.6 (−4.7 to 15.9)	−1.5 (−11.4 to 8.3)	
Independent	0.1 (−9.9 to 10.1)	−3.8 (−13.6 to 6.0)	0.3 (−9.8 to 10.4)	
Another party affiliation	−11.6 (−37.5 to 14.4)	4.1 (−21.8 to 30.0)	−2.3 (−28.6 to 24.0)	
Frequency of eating at restaurants				.84
Never or <1 time per week	−2.6 (−10.7 to 5.4)	−5.7 (−13.7 to 2.2)	−4.5 (−12.5 to 3.4)	
1-2 times per week	−2.3 (−10.3 to 5.6)	1.3 (−6.6 to 9.2)	−5.4 (−13.2 to 2.3)	
3-4 times per week	−9.0 (−22.6 to 4.6)	−5.4 (−19.0 to 8.3)	−5.7 (−19.4 to 8.0)	
5 times per week or more	5.6 (−22.4 to 33.6)	14.5 (−13.6 to 42.6)	10.7 (−17.2 to 38.6)	
Frequency of eating red meat				.68
1 time per week or less	−1.0 (−9.2 to 7.2)	.3 (−8.0 to 8.6)	−2.8 (−11.1 to 5.5)	
2-3 times per week	−4.5 (−12.8 to 3.8)	−4.3 (−12.6 to 3.9)	−5.9 (−14.0 to 2.1)	
4-6 times per week	−12.9 (−26.5 to .6)	−5.2 (−18.5 to 8.1)	−7.7 (−21.2 to 5.7)	
1 time per day or more	10.7 (−3.8 to 25.3)	2.9 (−11.8 to 17.6)	−1.9 (−16.4 to 12.6)	
Gender				.51
Female	−4.1 (−12.4 to 4.1)	−1.4 (−9.7 to 7.0)	−.5 (−8.8 to 7.7)	
Male	−3.3 (−9.9 to 3.3)	−1.9 (−8.5 to 4.6)	−6.9 (−13.5 to −0.4)	
Household poverty level				.29
≥150% federal poverty level	−5.1 (−11.4 to 1.2)	−4.6 (−10.9 to 1.7)	−4.3 (−10.6 to 2.0)	
<150% federal poverty level	1.1 (−7.8 to 10.0)	3.3 (−5.5 to 12.1)	−5.0 (−13.7 to 3.6)	

[a]*P* values are from Wald tests of significance of the coefficients on all interaction terms.
[b]ADE = average differential effect.
[c]GED = General Educational Development test.

Table 10. Impacts of red-meat–reduction messages on psychological outcomes, 2,773 US adults participating in the online randomized experiment evaluating red-meat–reduction messages[a]

Outcome	Impact of animal suffering message	Impact of health message	Impact of environmental message
	←ADE[b] (95% CI)→		
Message reactions			
Perceived message effectiveness	**.70[c] (.60 to .80)**	**.72[c] (.63 to .82)**	**.64[c] (.54 to .74)**
Negative emotions	**.51[c] (.41 to .61)**	**.43[c] (.33 to .53)**	**.43[c] (.33 to .53)**
Thinking about harms of red meat	**.72[c] (.62 to .82)**	**.67[c] (.58 to .77)**	**.70[c] (.60 to .80)**
Reactance	**.67[c] (.57 to .77)**	**.44[d] (.34 to .54)**	**.60[c] (.49 to .70)**
Perceptions			
Limiting red meat intake reduces harms	.10[c] (−.00001 to .21)	**.16[c] (.06 to .27)**	.09[c] (−.01 to .20)
Eating red meat contributes to harms	−.04[c] (−.14 to .06)	**.13[d] (.03 to .23)**	.003[c] (−.10 to .10)
Behavioral intentions			
Intentions to limit red meat intake	.09[c] (−.02 to .19)	.10[c] (−.001 to .21)	**.12[c] (.02 to .22)**

[a]Bold type indicates a statistically significant impact of the red-meat–reduction message compared with the control message ($P < .05$).
[b]ADE = average differential effect of red-meat–reduction message compared with the control message, as estimated from linear regressions. Dependent variables were standardized before analysis.
[cd]Within each row, ADEs with different superscript letters are different from one another ($P < .05$).